Right-Angled Triangles

Exam code: 9MA0

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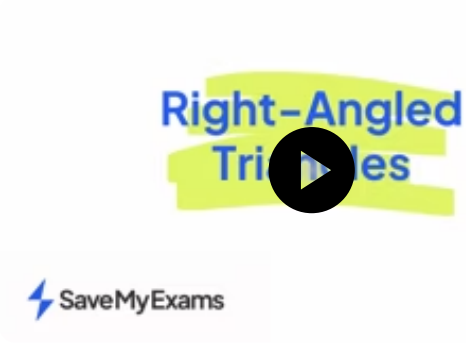


Did this video help you? Yes No

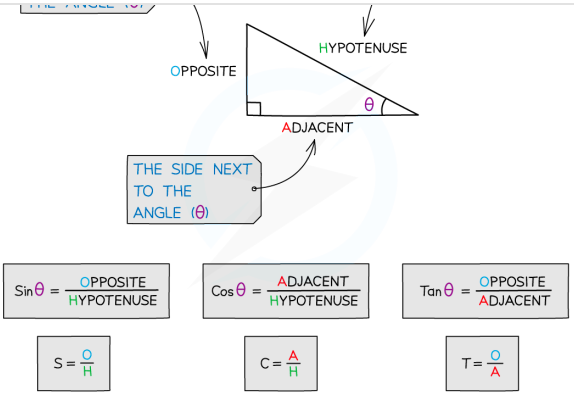
Help me understand

Right-angled triangles

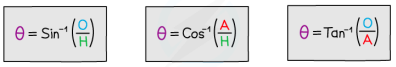
What is SOHCAHTOA?

- SOH, CAH, and TOA are an easy way to remember the three basic trigonometry ratios
- They show the relationship between an angle and the different sides of a right-angled triangle which we can label **Hypotenuse**, **Adjacent** and **Opposite**
- We use them to find missing angles and sides for right-angled triangles
  - **Hypotenuse** is the longest side of a right-angled triangle
  - **Adjacent** is the side next to the angle you are working with (in between the angle and the right angle)
  - **Opposite** is the side opposite the angle you are working with

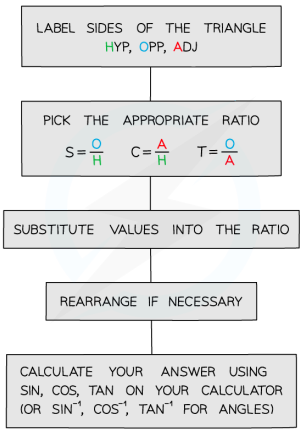
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- We use the inverse trig functions to find missing angles



How do I use SOHCAHTOA?

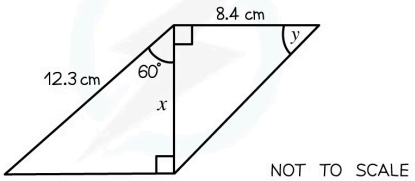


Examiner Tips and Tricks

- Remember SOHCAHTOA are only for calculating missing sides and angles for right-angled triangles
  - For non-right-angled triangles you will need to use the Sine or Cosine rules.
- Make sure your calculator is in degree mode (D).
- Always check your answers are sensible…
  - Should my answer be longer or shorter than the sides I have already i.e. is it the hypotenuse?
  - Is my angle sensible…
  - Have I remembered to use $^{-1}$?
- Check if the question asks you to round your answer.

Worked Example

Find the values of x and y in the following diagram.
Give your answers to 3 significant figures.



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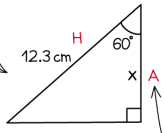
TO FIND x

SKETCH AND LABEL THE TRIANGLE



WE KNOW H AND WE WANT TO FIND A SO MUST NEED CAH

$\cos\theta = \frac{A}{H}$

$\cos 60 = \frac{x}{12.3}$

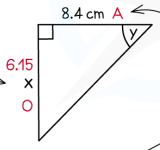
$x = 12.3 \times \cos 60$

SUBSTITUTE VALUES AND REARRANGE

$x = 6.15$ cm

TO FIND y

NOW WE'VE FOUND x=6.15, SKETCH AND LABEL THE TRIANGLE



WE HAVE A AND O SO MUST NEED TOA

$\tan\theta = \frac{O}{A}$

$\tan y = \frac{6.15}{8.4}$

$y = \tan^{-1}\left(\frac{6.15}{8.4}\right)$

USE $\tan^{-1}$ TO CALCULATE THE ANGLE

$y = 36.209...$

$y = 36.2$

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